

Computer security

Computer security (also **cybersecurity**, **digital security**, or **information technology (IT) security**) is the protection of **computer software**, **systems** and **networks** from **threats** that can lead to unauthorized information disclosure, theft or damage to **hardware**, software, or **data**, as well as from the disruption or misdirection of the **services** they provide.^{[1][2]}



An example of a physical security measure: a metal lock on the back of a personal computer to prevent hardware tampering.

The significance of the field stems from the expanded reliance on **computer systems**, the **Internet**,^[3] and **wireless network standards**. Its importance is further amplified by the growth of **smart devices**, including **smartphones**, **televisions**, and the various devices that constitute the **Internet of things** (IoT). Cybersecurity has emerged as one of the most significant new challenges facing the contemporary world, due to both the complexity of **information systems** and the societies they support. Security is particularly crucial for systems that govern large-scale systems with far-reaching physical effects, such as **power distribution**, **elections**, and **finance**.^{[4][5]}

Although many aspects of computer security involve digital security, such as electronic **passwords** and **encryption**, **physical security** measures such as **metal locks** are still used to prevent unauthorized tampering. IT security is not a perfect subset of **information security**, therefore does not completely align into the **security convergence** schema.

Vulnerabilities and attacks

A vulnerability refers to a flaw in the structure, execution, functioning, or internal oversight of a computer or system that compromises its security. Most of the vulnerabilities that have been discovered are documented in the [Common Vulnerabilities and Exposures](#) (CVE) database.^[6] An *exploitable* vulnerability is one for which at least one working [attack](#) or *exploit* exists.^[7] Actors maliciously seeking vulnerabilities are known as [threats](#). Vulnerabilities can be researched, reverse-engineered, hunted, or exploited using [automated tools](#) or customized scripts.^{[8][9]}

Various people or parties are vulnerable to cyber attacks; however, different groups are likely to experience different types of attacks more than others.^[10]

In April 2023, the [United Kingdom](#) Department for Science, Innovation & Technology released a report on cyber attacks over the previous 12 months.^[11] They surveyed 2,263 UK businesses, 1,174 UK registered charities, and 554 education institutions. The research found that "32% of businesses and 24% of charities overall recall any breaches or attacks from the last 12 months." These figures were much higher for "medium businesses (59%), large businesses (69%), and high-income charities with £500,000 or more in annual income (56%)."^[11] Yet, although medium or large businesses are more often the victims, since larger companies have generally improved their security over the last decade, [small and midsize businesses](#) (SMBs) have also become increasingly vulnerable as they often "do not have advanced tools to defend the business."^[10] SMBs are most likely to be affected by malware, ransomware, phishing, [man-in-the-middle attacks](#), and Denial-of Service (DoS) Attacks.^[10]

Normal internet users are most likely to be affected by untargeted cyberattacks.^[12] These are where attackers indiscriminately target as many devices, services, or users as possible. They do this using techniques that take advantage of the openness of the Internet. These strategies mostly include [phishing](#), [ransomware](#), [water holing](#) and scanning.^[12]

To secure a computer system, it is important to understand the attacks that can be made against it, and these [threats](#) can typically be classified into one of the following categories:

Backdoor

A [backdoor](#) in a computer system, a [cryptosystem](#), or an [algorithm](#) is any secret method of bypassing normal [authentication](#) or security controls. These weaknesses may exist for many

reasons, including original design or poor configuration.^[13] Due to the nature of backdoors, they are of greater concern to companies and databases as opposed to individuals.

Backdoors may be added by an authorized party to allow some legitimate access or by an attacker for malicious reasons. [Criminals](#) often use [malware](#) to install backdoors, giving them remote administrative access to a system.^[14] Once they have access, cybercriminals can "modify files, steal personal information, install unwanted software, and even take control of the entire computer."^[14]

Backdoors can be very hard to detect and are usually discovered by someone who has access to the application source code or intimate knowledge of the [operating system](#) of the computer.

Denial-of-service attack

[Denial-of-service attacks](#) (DoS) are designed to make a machine or network resource unavailable to its intended users.^[15] Attackers can deny service to individual victims, such as by deliberately entering a wrong password enough consecutive times to cause the victim's account to be locked, or they may overload the capabilities of a machine or network and block all users at once. While a network attack from a single [IP address](#) can be blocked by adding a new firewall rule, many forms of [distributed denial-of-service](#) (DDoS) attacks are possible, where the attack comes from a large number of points. In this case, defending against these attacks is much more difficult. Such attacks can originate from the [zombie computers](#) of a [botnet](#) or from a range of other possible techniques, including [distributed reflective denial-of-service](#) (DRDoS), where innocent systems are fooled into sending traffic to the victim.^[15] With such attacks, the amplification factor makes the attack easier for the attacker because they have to use little bandwidth themselves. To understand why attackers may carry out these attacks, see the 'attacker motivation' section.

Physical access attacks

A direct-access attack is when an unauthorized user (an attacker) gains physical access to a computer, most likely to directly copy data from it or steal information.^[16] Attackers may also compromise security by making operating system modifications, installing [software worms](#), [keyloggers](#), [covert listening devices](#) or using wireless microphones. Even when the system is protected by standard security measures, these may be bypassed by booting another

operating system or tool from a [CD-ROM](#) or other bootable media. [Disk encryption](#) and the [Trusted Platform Module](#) standard are designed to prevent these attacks.

Direct service attackers are related in concept to [direct memory attacks](#) which allow an attacker to gain direct access to a computer's memory.^[17] The attacks "take advantage of a feature of modern computers that allows certain devices, such as external hard drives, graphics cards, or network cards, to access the computer's memory directly."^[17]

Eavesdropping

[Eavesdropping](#) is the act of surreptitiously listening to a private computer conversation (communication), usually between hosts on a network. It typically occurs when a user connects to a network where traffic is not secured or encrypted and sends sensitive business data to a colleague, which, when listened to by an attacker, could be exploited.^[18] Data transmitted across an *open network* allows an attacker to exploit a vulnerability and intercept it via various methods.

Unlike [malware](#), direct-access attacks, or other forms of cyber attacks, eavesdropping attacks are unlikely to negatively affect the performance of networks or devices, making them difficult to notice.^[18] In fact, "the attacker does not need to have any ongoing connection to the software at all. The attacker can insert the software onto a compromised device, perhaps by direct insertion or perhaps by a virus or other malware, and then come back some time later to retrieve any data that is found or trigger the software to send the data at some determined time."^[19]

Using a [virtual private network](#) (VPN), which encrypts data between two points, is one of the most common forms of protection against eavesdropping. Using the best form of encryption possible for wireless networks is best practice, as well as using [HTTPS](#) instead of an unencrypted [HTTP](#).^[20]

Programs such as [Carnivore](#) and [NarusInSight](#) have been used by the [Federal Bureau of Investigation](#) (FBI) and NSA to eavesdrop on the systems of [internet service providers](#). Even machines that operate as a closed system (i.e., with no contact with the outside world) can be eavesdropped upon by monitoring the faint [electromagnetic](#) transmissions generated by the hardware. [TEMPEST](#) is a specification by the NSA referring to these attacks.